Worksheet 3

Instructions

- Include your full name and Coyote ID on the top right of the first page.
- Submit completed worksheets to Gradescope (no tagging required).
- Submit before 11:59 PM on Wednesday, September 16, 2026

Physics Problem Solving Tips

- Draw a diagram (if possible).
- Write down what you know.
- Write down what you want to find.
- First solve symbolically; plug in values at the end.
- Check to make sure your units are correct.
- If you plug in values, check order of magnitude.

Problems

This worksheet covers kinematics in 1D, free fall, and incline planes.

1. A skier is gliding along at $3.0\ \mathrm{m/s}$ on horizontal, frictionless snow. He suddenly starts down a 10° incline. His speed at the bottom is $15\ \mathrm{m/s}$.
   a. What is the length of the incline?
   b. How long does it take him to reach the bottom?

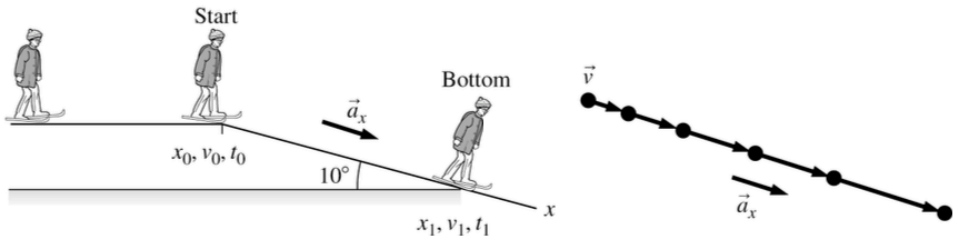
MODEL: We will model the skier as a particle.

VISUALIZE:

Known
$x_0 = 0 \quad t_0 = 0$
$v_0 = 3.0$ m/s
$a_x = g \sin 10°$
$v_1 = 15$ m/s

Find
$x_1 \quad t_1$

Pictorial representation



Note that the skier's motion on the horizontal, frictionless snow is not of any interest to us. Also note that the acceleration parallel to the incline is equal to $g \sin 10°$.

SOLVE: **(a)** Using the following constant-acceleration kinematic equations,

$$v_{fx}^2 = v_{ix}^2 + 2a_x(x_f - x_i)$$
$$\Rightarrow (15\ \mathrm{m/s})^2 = (3.0\ \mathrm{m/s})^2 + 2(9.8\ \mathrm{m/s^2})\sin 10°(x_1 - 0\ \mathrm{m}) \Rightarrow x_1 = 64\ \mathrm{m}$$
$$v_{fx} = v_{ix} + a_x(t_f - t_i)$$
$$\Rightarrow (15\ \mathrm{m/s}) = (3.0\ \mathrm{m/s}) + (9.8\ \mathrm{m/s^2})(\sin 10°)t \Rightarrow t = 7.1\ \mathrm{s}$$

(b) Using a different kinematic equation this time, we find

$$v_{fx} = v_{ix} + a_x(t_f - t_i)$$
$$\Rightarrow (15\ \mathrm{m/s}) = (3.0\ \mathrm{m/s}) + (9.8\ \mathrm{m/s^2})(\sin 10°)t \Rightarrow t = 7.1\ \mathrm{s}$$

REVIEW: A time of 7.1 s to cover 64 m is a reasonable value.

2. A skateboarder starts up a $3.0\ \text{m}$ long ramp at $4.0\ \text{m/s}$. What is the maximum height of the ramp where the skateboarder goes off the end rather than rolling back down?

MODEL: We will model the skateboarder as a particle undergoing constant acceleration.

VISUALIZE: Because the skateboarder is on an incline, we can use equation 2.26 to describe his/her acceleration. Note that $\sin(\theta) = h/\Delta x$, where h is the vertical height we seek, and Δx is the distance along the incline. Because we are seeking the highest possible incline, we will consider the case where the speed has dropped almost to zero, and the skateboarder just barely makes it to the top. We can use $v_{0x}^2 = v_{fx}^2 + 2a_x\Delta x$.

Solve: Combining the expressions above, we have

$$v_{fx}^2 = v_{0x}^2 + 2a_x\Delta x = v_{0x}^2 - 2g\sin(\theta)\Delta x = v_{0x}^2 - 2g\frac{h}{\Delta x}\Delta x$$

$$v_{fx}^2 = v_{0x}^2 - 2gh \Rightarrow h = \frac{v_{fx}^2 - v_{0x}^2}{-2g}$$

$$h = \frac{(0\ \text{m/s})^2 - (4.0\ \text{m/s})^2}{-2(9.8\ \text{m/s}^2)} = 0.82\ \text{m}$$

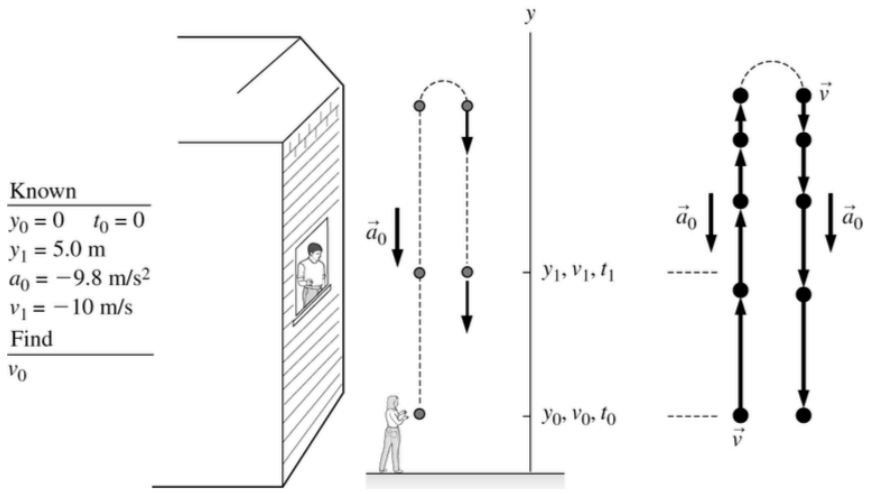
REVIEW: The initial speed of 4.0 m/s is a plausible speed for a skateboarder. So, we expect an intuitively plausible height for the skateboarder to reach going up a ramp. A height just under 1 m seems reasonable.

3. Nicole throws a ball straight up. Chad watches the ball from a window $5.0\,\mathrm{m}$ above the point where Nicole released it. The ball passes Chad on the way up, and it has a speed of $10\,\mathrm{m/s}$ as it passes him on the way back down. How fast did Nicole throw the ball?

MODEL: The ball is a particle that exhibits freely falling motion according to the constant-acceleration kinematic equations.

VISUALIZE:

Pictorial representation



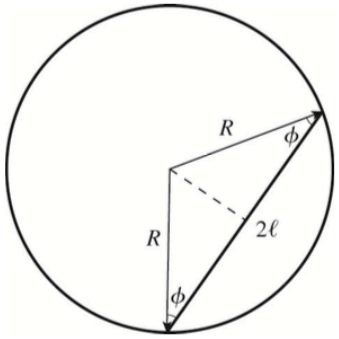
SOLVE: Using the known values, we have

$$v_1^2 = v_0^2 + 2a_0(y_1 - y_0) \Rightarrow (-10\ \mathrm{m/s})^2 = v_0^2 + 2(-9.8\ \mathrm{m/s^2})(5.0\ \mathrm{m} - 0\ \mathrm{m}) \Rightarrow v_0 = 14\ \mathrm{m/s}$$

4. Figure 1 shows a fixed vertical disk of radius R. A thin frictionless rod is attached to the bottom point of the disk and to a point on the edge, making angle ϕ with the vertical. Find an expression for the time it takes a bead to slide from the top end of the rod to the bottom

MODEL: We can model the bead as a particle undergoing constant acceleration.

VISUALIZE: Because this bead is effectively on an incline, we can use equation 2.26. Note that the angle used in equation 2.26 is measured from horizontal, such that we will use $a_x = \pm g\sin(90° - \phi) = \pm g\cos(\phi)$. In this case, we will choose our axes such that $+x$ points up the incline, and $a_x = -g\cos(\phi)$. We can use kinematic equations, but we will need a total distance traveled. Refer to the diagram below.



Note that $R\cos(\phi) = \ell$, such that the total distance the bead must traverse is $2\ell = 2R\cos(\phi)$. With this distance, we can apply $\Delta x = v_{0x}\Delta t + \frac{1}{2}a_x(\Delta t)^2$.

SOLVE: Combining the above expressions, and using the fact that the bead starts from rest, we have

$$\Delta x = -2R\cos(\phi) = (0)\Delta t + \frac{1}{2}(-g\cos(\phi))(\Delta t)^2$$

$$\Delta t = \sqrt{\frac{4R}{g}} = 2\sqrt{\frac{R}{g}}$$